

0 4 . 1

Binding of one molecule of oxygen to haemoglobin makes it easier for a second oxygen molecule to bind.

Explain why.

[2 marks]

Question 4 continues on the next page

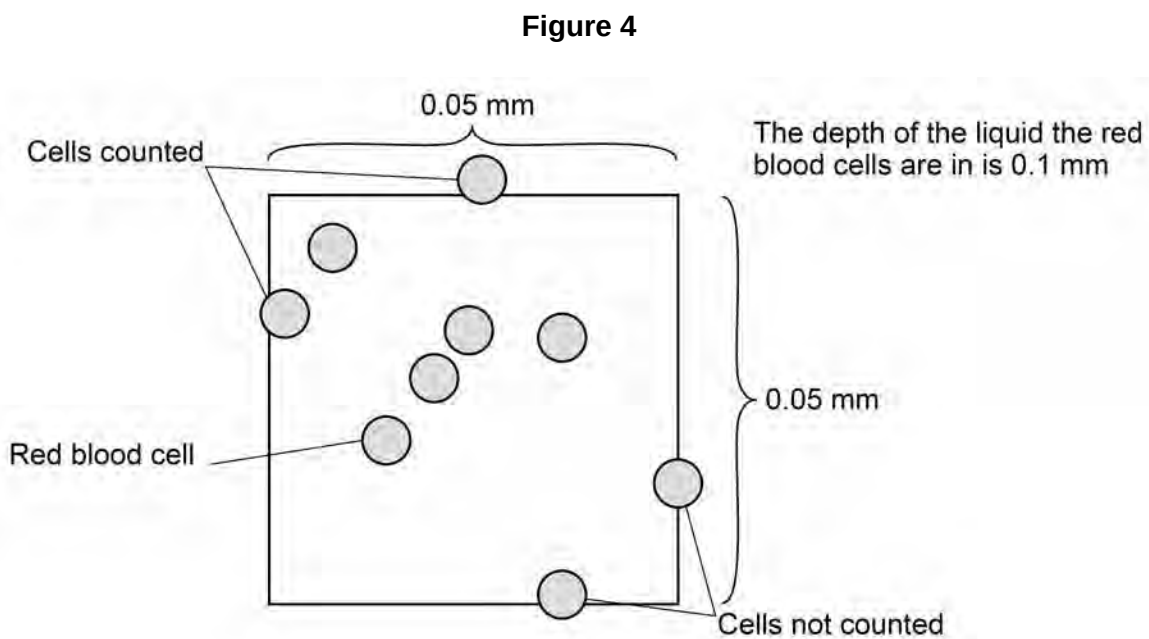
Turn over ►

A haemocytometer is a special microscope slide which can be used to count the numbers of blood cells in a sample of blood.

- The surface of the slide has many small, equal-sized squares marked on it.
- The depth of the liquid under each square is 0.1 mm
- When counting, cells that touch top or left lines are counted but cells that touch right or bottom lines are not counted.

A doctor used a haemocytometer to determine the number of red blood cells per mm^3 in a blood sample. He diluted the original blood sample by a factor of 200 times before putting some on a haemocytometer.

Figure 4 shows the distribution of cells in a typical small square.



0 4 . 2

The doctor counted the red blood cells in many small squares.
The **mean** number of red blood cells per small square was 7
The original blood sample was diluted by a factor of 200 times.

Calculate the number of red blood cells per mm^3 in the original blood sample.
Give your answer in standard form.

[2 marks]

Answer = _____ red blood cells per mm^3

0 4 . 3

When counting, cells that touch top or left lines are counted but cells that touch right or bottom lines **are not** counted.

Suggest **two** reasons for this rule.

[2 marks]

1 _____

2 _____

Turn over ►

The doctor also wanted to know how many white blood cells per mm^3 there were in a different sample of blood. To do this he first diluted the sample by a factor of 20 times. He then made the white blood cells clearly visible by using a stain that makes nuclei appear dark blue.

0 4 . 4

When counting white blood cells, the doctor only diluted the blood sample by a factor of 20 times, instead of 200 times when counting red blood cells.

Suggest why he only diluted the sample by a factor of 20 times.

[1 mark]

0 4 . 5

Explain how the stain allowed the doctor to count the white blood cells amongst all the red blood cells.

[1 mark]

